

Analytical MCQ Assessment

High-level cognitive evaluation paper focusing on firewalls and their types.

SECTION A: Multiple Choice Questions

Q1. An administrator wants to block incoming traffic based solely on source IP and port number without tracking the state of active connections. Which type of firewall is most suitable for this task? (1 Marks)

- A. Stateful Inspection Firewall
- B. Packet Filtering Firewall
- C. Application-Level Gateway
- D. Next-Generation Firewall

Q2. How does a stateful inspection firewall determine whether to allow an incoming packet that does not match an explicit rule? (1 Marks)

- A. It performs deep packet inspection on the payload.
- B. It checks if the packet is part of an established connection in its state table.
- C. It routes the packet to a proxy server for verification.
- D. It automatically blocks all incoming packets by default regardless of state.

Q3. Which type of firewall operates at the Application Layer (Layer 7) of the OSI model and acts as an intermediary by terminating the incoming connection and initiating a new one to the destination? (1 Marks)

- A. Circuit-Level Gateway
- B. Packet Filtering Firewall
- C. Proxy Firewall
- D. Stateful Inspection Firewall

Q4. A network engineer deploys a firewall that validates TCP handshakes and monitors session initiation without inspecting the actual packet payloads. What type of firewall is this? (1 Marks)

- A. Circuit-Level Gateway
- B. Application Firewall
- C. Packet Filter
- D. Next-Generation Firewall

Q5. What primary feature distinguishes a Next-Generation Firewall (NGFW) from a traditional stateful firewall? (1 Marks)

- A. Ability to filter traffic based on MAC addresses.
- B. Integrated deep packet inspection, application awareness, and intrusion prevention.
- C. Support for basic NAT and VPN tunneling.
- D. Faster packet routing at the physical layer.

Q6. A company needs to protect its web servers specifically from SQL injection and cross-site scripting (XSS) attacks. Which specialized firewall should they deploy? (1 Marks)

- A. Stateful Inspection Firewall
- B. Web Application Firewall (WAF)
- C. Circuit-Level Gateway
- D. Packet Filtering Router

Q7. Which of the following best describes a key advantage of a host-based firewall compared to a network-based firewall? (1 Marks)

- A. It protects the entire network segment from external threats.
- B. It can inspect encrypted traffic before it is decrypted by the host.
- C. It provides tailored protection for a specific device, including internal process monitoring.
- D. It requires no configuration or maintenance from system administrators.

Q8. In a secure network architecture, where should public-facing servers (such as web and mail servers) be placed to isolate them from the internal private network? (1 Marks)

- A. Inside the intranet segment
- B. Directly on the public Internet
- C. Within a Demilitarized Zone (DMZ)
- D. Behind a simple packet-filtering router

Q9. When securing a multi-tenant cloud environment, which firewall deployment model is best suited for filtering east-west traffic between virtual machines on the same physical host? (1 Marks)

- A. Hardware-based appliance
- B. Virtual Firewall
- C. Circuit-Level Gateway
- D. Bastion Host

Q10. Which of the following security threats is a traditional packet-filtering firewall least capable of detecting? (1 Marks)

- A. An unauthorized connection attempt to a closed port.
- B. A spoofed IP address originating from an external network.
- C. A malicious executable file hidden within an allowed HTTP download.
- D. Traffic originating from an unapproved source IP address.

INSTRUCTOR ONLY - ANSWER KEY & RUBRIC

MCQ Solutions & Rationale

Q1. Correct Answer: B

Rationale: Packet filtering firewalls inspect packets individually based on header information like IP addresses and ports without tracking connection state.

Q2. Correct Answer: B

Rationale: Stateful firewalls keep track of active connections in a state table to allow return traffic for established sessions.

Q3. Correct Answer: C

Rationale: A proxy firewall acts as an intermediary, establishing separate connections to the client and the server to inspect application-layer data.

Q4. Correct Answer: A

Rationale: Circuit-level gateways operate at the Session layer to verify TCP handshakes and session validity without inspecting payload content.

Q5. Correct Answer: B

Rationale: NGFWs combine traditional firewall capabilities with deep packet inspection, application-level control, and integrated intrusion prevention systems.

Q6. Correct Answer: B

Rationale: A Web Application Firewall (WAF) is designed specifically to filter and monitor HTTP/HTTPS traffic to protect web applications from exploits like SQL injection and XSS.

Q7. Correct Answer: C

Rationale: Host-based firewalls run directly on individual devices, allowing them to monitor local processes and provide customized protection for that specific host.

Q8. Correct Answer: C

Rationale: A Demilitarized Zone (DMZ) isolates public-facing services from the internal network, limiting exposure if a public server is compromised.

Q9. Correct Answer: B

Rationale: Virtual firewalls run as software appliances within virtualized environments to monitor and secure traffic between virtual machines.

Q10. Correct Answer: C

Rationale: Traditional packet-filtering firewalls only inspect packet headers and cannot analyze the payload for malicious content like malware.